Лабораторная работа №1

Вариант №3

Листинг библиотеки классов (dll):

```
using System;

namespace dll
{
    public class Machine
    {
        public string Metal { get; set; } = "None";
        public int Age { get; set; } = 0;
        public bool Break { get; set; } = false;

        public void Repair()
        {
            if (Break)
            {
                Break = false;
                Console.WriteLine("Car is fixed");
            }

            else
                Console.WriteLine("Car is in good condition");

        }

        public void setAge(int new_age)
        {
            Age = new_age;
            Console.WriteLine($"Set age: {Age}");
        }

        public void ShowInfo()
        {
            string Status = Checking();
            Console.WriteLine($"\nMachine age: {Age}\nStatus: {Status}\nMetal
    : {Metal}");
        }

        public string Checking()
        {
            if (!Break)
                return "serviceable";
            else
                return "break";
        }
    }

    public class Car : Machine
    {
        public int id { get; set; } = 0;
        public string Model { get; set; } = "None";
        public int Places = 0;
        public Engine engine = new Engine();
```

```
        public bool Drive = false;


        public void DriveToggle()
        {
            if (Drive) { Drive = false; }
            else { Drive = true; }

            if (Drive && !Break)
            {


                if (!engine.Work)
                {
                    engine.Start();
                }

                Console.WriteLine("Drive");
            }

            else if (Break)
            {
                Drive = false;
                Console.WriteLine("Car is broken!");
            }
            else
            {
                if (engine.Work)
                {
                    engine.Stop();
                }
                Console.WriteLine("Stop drive");
            }
        }

        public void Honk() { Console.WriteLine("HONK!"); }

        public void ReplaceEngine()
        {
            if (engine.Work) { engine.Stop(); }

            Console.WriteLine("Model engine: ");
            string model = Console.ReadLine();

            Console.WriteLine("Cylinders engine: ");
            int cylinders = int.Parse(Console.ReadLine());

            Console.WriteLine("HorsePower engine: ");
            int horse_power = int.Parse(Console.ReadLine());

            Engine new_engine = new Engine { model_engine = model, Cylinders = cylinders, HorsePower = horse_power };
            engine = new_engine;

        }

```

```
        public void modelEngine()
        {
            Console.WriteLine(engine.model_engine);

        }

        public Car CreateCar()
        {
            Car newCar = new Car();

            Console.WriteLine("Model: ");
            newCar.Model = Console.ReadLine();

            Console.WriteLine("Places: ");
            newCar.Places = int.Parse(Console.ReadLine());

            Console.WriteLine("--Machine Properties--");
            Console.WriteLine("Metal: ");
            newCar.Metal = Console.ReadLine();

            Console.WriteLine("Age: ");
            newCar.Age = int.Parse(Console.ReadLine());

            Console.WriteLine("Break (true/false): ");
            newCar.Break = bool.Parse(Console.ReadLine());

            Console.WriteLine("--Engine Properties--");
            /*newCar.engine = new Engine();*/
            newCar.engine.createEngine();

            return newCar;

        }

    }

    public class Engine
    {
        public int id { get; set; } = 0;
        public string model_engine { get; set; } = "None";
        public int Cylinders { get; set; } = 0;
        public int HorsePower { get; set; } = 0;
        public bool Work = false;

        public Engine createEngine()
        {

            Console.WriteLine("Model: ");
            string Model_engine = Console.ReadLine();
            Console.WriteLine("Cylinders: ");
            int cylinders = int.Parse(Console.ReadLine());
            Console.WriteLine("HorsePower: ");
            int horsePower = int.Parse(Console.ReadLine());

            return new Engine { Cylinders = Cylinders, HorsePower = 
    horsePower, model_engine = Model_engine };
```

```
        }

        public void Start()
        {
            if (!Work) { Work = true; } else { Console.WriteLine("The engine is already running"); }
        }

        public void Stop()
        {
            if (Work) { Work = false; } else { Console.WriteLine("The engine is no longer running"); }
        }


        public void Upgrade(int extraHorsPower)
        {
            HorsePower = extraHorsPower;
        }

    }
}
```

Листинг консольного приложения

```
using dll;

class Program
{
    static public void Main()
    {
        bool exit = false;

        do
        {
            Console.Clear();
            Console.WriteLine("1. Create car");
            Console.WriteLine("2. Info car");
            Console.WriteLine("0. Exit");
            int choice = int.Parse(Console.ReadLine());

            switch (choice)
            {
                case 1:
                    Car newCar = new Car();
                    newCar.CreateCar();
                    break;
                case 0:
                    exit = true;
                    Console.Clear();
                    break;
            }

        } while (!exit);
    }
}
```

Результаты работы программы: